

Tyler Johnston

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PROFESSIONAL SUMMARY

Associate Back-end Engineer at WPP Enterprise Solutions, currently working with AWS and preparing for TypeScript and AI systems work in the e-commerce sector. MSc Information Management candidate at NOVA IMS, specializing in Business Intelligence. Experience across production application development, REST APIs, databases, computer vision research, analytics, QA, and production support.

EDUCATION

NOVA IMS, Lisbon, Portugal

Sep 2025 - Feb 2027

Master's in Information Management | Business Intelligence Specialization | Currently Enrolled

Utah State University, Logan, Utah (USA)

Aug 2020 - May 2024

B.S. in Computer Science, Minor in Mathematics | Magna Cum Laude | GPA: 3.92 / 4.00 | Presidential Scholarship

SKILLS

Programming C#, JavaScript, TypeScript, Python, Java, C++, Oracle SQL, PostgreSQL

Frameworks & Tools .NET, REST APIs, React.js, Angular, Docker, Firebase, AWS, Power BI, Microsoft Fabric, Supabase

DevOps & QA Git, Agile/Scrum, CI/CD, debugging, application testing, QA, production support

Languages English (Native), Portuguese (B1)

EXPERIENCE

Associate Back-end Engineer, WPP Enterprise Solutions | Porto, Portugal

Sep 2026 - Present

- Working with AWS during initial onboarding, with planned back-end development in TypeScript and AI systems work for the e-commerce sector.

Full Stack Developer, Footprints, Inc. (aerospace/manufacturing software) | Logan, UT

Mar 2024 - Aug 2025

- Built and maintained production workflow applications for aerospace and steel-manufacturing engineering teams using C#/.NET, REST APIs, JavaScript, Ext JS, jQuery, Oracle SQL, 4D, and Visual Basic.
- Completed 240+ full-stack development tasks across Agile sprints, translating stakeholder requirements into front-end, back-end, database, API, and workflow automation features.
- Served as a dedicated QA owner for full-stack releases, testing front-end, back-end, API, and database changes before deployment and resolving defects through root-cause analysis.
- Optimized SQL queries, service-layer logic, and data retrieval workflows to improve reliability, maintainability, and performance across production systems.

Computer Vision Research Assistant, Utah State University | Logan, UT

Jan 2024 - May 2024

- Built computer vision pipelines using YOLOv8 and Roboflow to detect curb ramps, pedestrian signage, and traffic lights in Google Street View imagery for ADA compliance analysis.
- Automated image retrieval, GPS metadata extraction, and data preprocessing workflows to create geospatial datasets and heatmaps for city-scale accessibility evaluation.

Modern C++ Teaching Assistant, Utah State University | Logan, UT

Aug 2023 - Dec 2023

- Designed a custom C++ std::vector prototype using templates, iterators, and manual memory management to optimize large-dataset allocation and demonstrate advanced data structure design.
- Held weekly office hours and graded assignments for an upper-division C++ course, providing debugging support on memory management, templates, iterators, and STL containers.

Web Developer, Space Dynamics Laboratory | Remote

Mar 2023 - Jun 2023

- Developed a full-stack workflow design tool using React, React Flow, C#/.NET, and Docker microservices, enabling users to create process workflows through a responsive drag-and-drop interface instead of manual XML editing.
- Enabled JSON and XML workflow configuration import/export through Docker-integrated services, reducing manual configuration errors and improving developer productivity.

PROJECTS

Flashcard Dungeon | Full-Stack Study App (flashcarddungeon.com)

Angular / TypeScript / Supabase

- Built a general-purpose Angular/TypeScript study app pairing FSRS-6 spaced repetition with RPG combat, unlockable characters, a monster bestiary, deck notes, and multiple study modes.
- Implemented a NOVA IMS MSc thesis platform prepared for a pilot study, with a versioned study protocol and randomized supplementary retrieval of previously reviewed, non-due cards sampled without replacement.
- Built an append-only IndexedDB research ledger with idempotent Supabase/PostgreSQL sync; gated FSRS updates on single-use review authorizations and logged supplementary retrieval without rescheduling.

Parallax Poker | Character-Driven Poker Roguelite

C# / Godot 4 / Python

- Built a C# / Godot 4 Texas Hold'em roguelite with a rules-correct poker engine covering all-ins, raise/reopening logic, uncalled chip refunds, O(1) hand evaluation, and asynchronous game-state management.
- Developed a personality-driven AI decision engine with 15+ behavior parameters, equity calculations, draw detection, board texture scoring, SPR logic, tilt states, and probabilistic betting decisions.
- Created a Monte Carlo balancing pipeline using headless Godot simulations, Python, Pandas, and Plotly to test 9 AI opponents across 288,000 simulated hands and identify win-rate outliers.

Absentee Analysis | HR Absenteeism Dashboard

Python / Pandas

- Built a Vercel-hosted HR analytics tool that segmented employees into absenteeism risk groups using clustering models and a Random Forest classifier, enabling non-technical HR users to review risk segments and support data-driven workforce decisions.